Derek Lund

B.S. Aerospace Engineering | UCLA | Expected 2028

Aerospace engineering student specializing in CFD-based aerodynamic analysis and high-power rocketry. Aerodynamics Lead for the OCC Rocketry Club IREC competition team, leading geometry optimization through simulation and wind tunnel validation. L1 high-power rocketry certified.

FOCUS AREAS

Aerodynamic Simulation

ANSYS Fluent (SST k-omega) CFD analysis of rocket geometries at subsonic and transonic conditions. Parametric studies comparing axial force, normal force, and pitching moment across multiple designs and flight conditions.

Experimental Validation

Design and fabrication of modular 3D-printed subscale rocket models for wind tunnel force measurements. Planned testing at Cal Poly Pomona to validate CFD predictions against experimental data.

Vehicle Design & CAD

Full-scale IREC competition rocket design in OnShape, translating CFD-optimized geometry into dimensioned manufacturing drawings. Includes fin assembly, boattail, and nose cone components with tolerances.

High-Power Rocketry

Tripoli/NAR Level 1 certified. Constructed and flew LOC Precision Hi-Tech rocket from kit, developing hands-on skills in structural bonding, motor systems, and recovery deployment.

TOOLS & SKILLS

Simulation

ANSYS Fluent 2025 R2 | SST k-omega | Mesh Generation

CAD

OnShape | OpenRocket | Parametric Modeling

Fabrication

3D Printing | Epoxy Bonding | Subscale Model Build

Rocketry

High-Power Rocketry (L1) | LOC Precision | Recovery Systems

PROJECT 01 | CFD AERODYNAMIC ANALYSIS — IREC ROCKET GEOMETRY STUDY

Comparative CFD Study: Nose Cone & Fin Geometry Optimization

ANSYS Fluent 2025 R2 | SST k-omega | Mach 0.6 & 0.9 | AoA 0 to 10 deg

Role	Lead CFD Analyst — OCC Rocketry Club
Tools	ANSYS Fluent 2025 R2, SST k-omega turbulence model, ANSYS Meshing
Scope	4 designs (BTTE1, BTFE2, cDr3, T1.75BN); 1.5 in diameter subscale; Mach 0.6 & 0.9; AoA 0-10 deg
Domain	3 x 3 x 9 m enclosure; first inflation layer 1.8e-6 m; y+ < 1 target
Goal	Minimize axial drag while maintaining stability for IREC 10,000 ft altitude target

Note: Simulations performed under the ANSYS Student license (1.04M element limit, 10^-3 continuity convergence). Higher-fidelity simulations are planned for the coming months as the team scales up to the full-size IREC vehicle.

Methodology

- Geometry variants created in OpenRocket (varying fin span, root/tip chord, nose cone profile, and boattail), then modeled as full CAD in OnShape at 1.5 in subscale diameter.
- Identical mesh strategy applied to all four designs for a controlled comparison: unstructured hybrid mesh with dense inflation layers at nose and fin surfaces.
- Collected axial force FA, normal force FN, and pitching moment coefficient CM at Mach 0.6 and 0.9 across AoA 0-10 deg in 2.5 deg increments.
- Solution quality verified via scaled residuals and wall y+ contours confirming y+ =< 1 over the full geometry surface.

Results — Axial Force FA at 0 deg AoA

Design	FA at Mach 0.6 (lb)	FA at Mach 0.9 (lb)
BTTE1	**2.22**	**4.80**
BTFE2	2.45	5.36
cDr3	3.10	6.86
T1.75BN	2.33	5.09

Selected Design: BTTE1

BTTE1 achieved the lowest axial force at both Mach numbers — 2.22 lb at Mach 0.6 and 4.80 lb at Mach 0.9. The highest-drag design (cDr3) produced roughly 40% greater FA. T1.75BN was competitive on drag but showed less favorable FN and CM behavior across the AoA sweep. BTTE1 was selected as the baseline geometry for the IREC competition vehicle.

Parametric Study — FA, FN, and CM vs. Angle of Attack

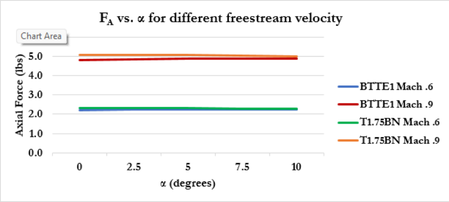

Fig. 1 — FA vs. AoA

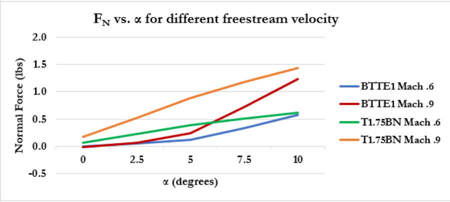

Fig. 2 — FN vs. AoA

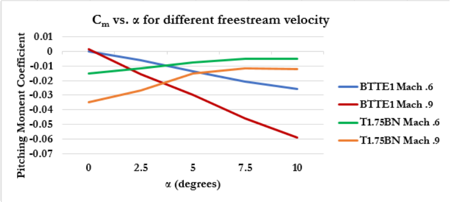

Fig. 3 — CM vs. AoA

Axial force (FA) is nearly independent of angle of attack for BTTE1 and T1.75BN, confirming drag is dominated by form drag. Integrated across the 0–10° AoA sweep, BTTE1 produced 12.2% and 46.5% lower cumulative normal force (FN) than T1.75BN at Mach 0.6 and Mach .9, respectively. The pitching moment coefficient (CM) trends in opposite directions for the two designs—a behavior that will be investigated further in higher-fidelity simulations.

PROJECT 01 (CONT.) | FLOW VISUALIZATION & SOLUTION QUALITY

Velocity Flow — BTTE1 at 0 deg AoA, Mach 0.9

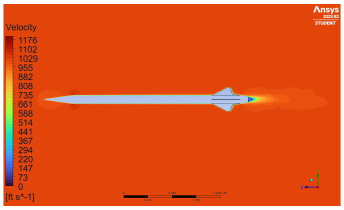

Fig. 4 — Velocity contour at 0 deg AoA, Mach 0.9.

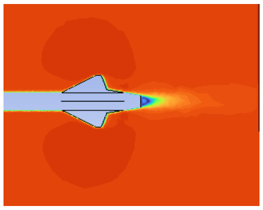

Fig. 5 — Velocity contour of fins at 0 deg AoA, Mach 0.9.

The velocity field shows attached flow along the airframe. Flow decelerates to a stagnation point at the nose tip before accelerating along the conical surface. At 0 deg AoA, symmetry in the wake structure aft of the fins and boattail is evident, consistent with the normal force values observed in the parametric study. The low-velocity recirculation zone immediately aft of the boattail is the primary source of base drag. However, the wake remains concentrated aft of the boattail rather than separating near the fin root, suggesting the fin geometry promotes smoother aft pressure recovery.

Surface Pressure Distribution — BTTE1 at Mach 0.9

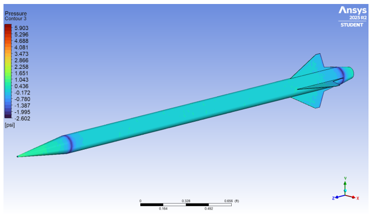

Fig. 6 — Full vehicle pressure contour at Mach 0.9, 0 deg AoA (psi).

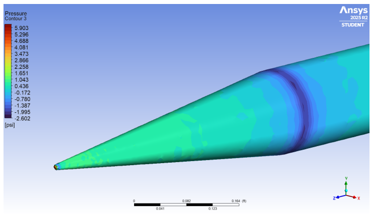

Fig. 7 — Pressure contour, nose cone close-up.

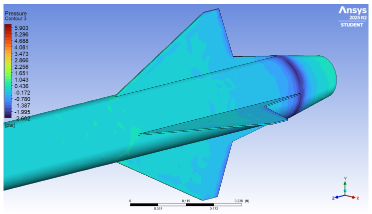

Fig. 8 — Pressure contour, fin close-up.

The surface pressure distribution at Mach 0.9 shows the conical nose cone develops a high-pressure stagnation region at the tip that transitions smoothly to near-freestream pressure along the body, with no adverse pressure gradient that would indicate flow separation. The close-up of the fin and boattail section reveals a low-pressure region on the upper fin surfaces and across the boattail transition, contributing significantly to the measured normal force. The relatively uniform cyan distribution across the fuselage confirms that body form drag is low, with the majority of the axial force attributable to the fin and boattail geometry.

Solution Quality — Mesh, Convergence & Wall y+

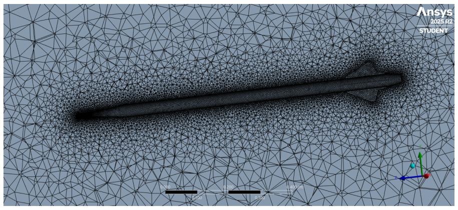

Fig. 9 — Mesh for BTTE1

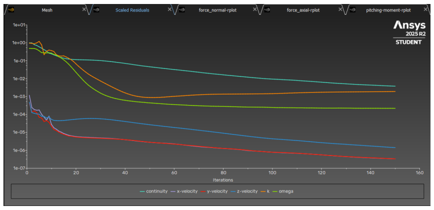

Fig. 10 — Scaled residuals for BTTE1.

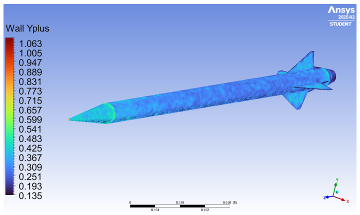

Fig. 11 — Wall y+ at Mach 0.6.

To determine mesh quality, we consider wall y+ values as well as scaled residual values. Wall y+ values are seen to be < .5 over the body of the rocket, with a maximum value of ~1 at the nose cone stagnation point. Our dense inflation layers allow us to accurately collect force and moment data. The scaled residuals all converge to an order of at most 10^-3. These metrics indicate adequate boundary-layer resolution and solver convergence. With the student license, we are limited to a 1.04M element limit and a 10^-3 convergence limit. Higher-fidelity meshing with full ANSYS license access will enable us to improve simulation accuracy with motor exhaust and a higher-detail CAD file, including attributes fin edge chamfer and fasteners, with more elements for the required inflation layer.

PROJECT 02 | WIND TUNNEL VALIDATION — SUBSCALE MODULAR MODEL

3D-Printed Subscale Model for Cal Poly Pomona Wind Tunnel Testing

Modular Design | 3D Printing | Experimental Aerodynamics | CFD Validation

Role	Design Lead — subscale model design and fabrication
Facility	Cal Poly Pomona subsonic wind tunnel
Model	Modular 3D-printed subscale; swappable fin sets and removable boattail for multi-config testing
Purpose	Experimental validation of CFD force predictions (FA, FN, CM) across AoA sweep
Status	Model fabricated and assembled; wind tunnel runs scheduled for end of May 2026

Wind tunnel data collection is ongoing — runs scheduled for completion by end of May 2026. Quantitative force measurements will be added to this portfolio once experimental results are available.

Design Philosophy

A key challenge in comparative wind tunnel testing is maintaining identical conditions across multiple design variants. The subscale model was designed so that fin sets are fully swappable and the boattail section and nose cone can be added or removed without rebuilding the fuselage. This allows back-to-back comparison runs under consistent tunnel conditions and sensor mounting, directly corresponding to the four designs evaluated in CFD.

Fabrication & Assembly

- All components 3D-printed in-house; wall thicknesses and interfaces toleranced for repeatable assembly and secure force-sensor mounting.
- Fin sets are individually swappable, allowing each geometry variant to be tested under identical fuselage and mounting conditions. Boattail section can be attached or removed independently to isolate its aerodynamic contribution from the fin effects.
- Internal structure designed to accommodate the tunnel force balance sensor rod through the fuselage centerline.
- Fuselage OD scaled to 41 mm to fit commercially available rocket tube stock, simplifying production and ensuring a consistent outer mold line.
- Initial assembly revealed the fuselage requires reinforcement at the sensor mount point — a redesign with an aluminum insert is planned for the next iteration.

Fabricated Components & Internal CAD

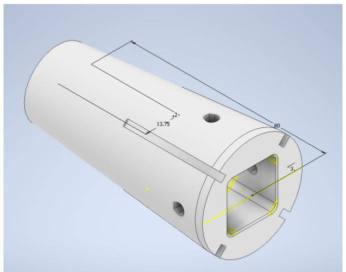

Fig. 12 — Removable boattail section CAD.

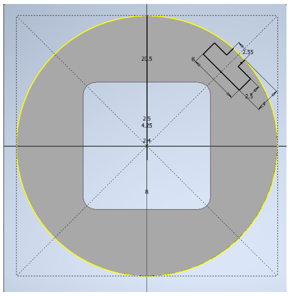

Fig. 13 — Cross-section view of internal geometry.

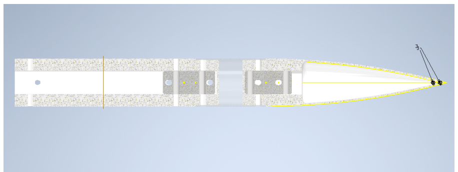

Fig. 14 — Nose cone section CAD side profile.

Fig. 15 — Disassembled 3D-printed subscale components.

Fig. 16 — Assembled subscale of BTTE1.

PROJECT 03 | IREC COMPETITION ROCKET — 3D DESIGN

BTTE1 Full-Scale IREC Vehicle — OnShape CAD Design

OnShape | Parametric CAD | Engineering Drawings | IREC Competition

Role	Aerodynamics Lead & CAD Designer — OCC Rocketry Club Competition Team
Tools	OnShape; geometry driven by ANSYS Fluent CFD results from Project 01
Vehicle	Approx. 4 m length, 6 in (152 mm) diameter, 4.4 lb payload, 10,000 ft target altitude
Status	In progress— major components modeled; dimensions and tolerances being finalized

Overview

Based on the CFD study selecting BTTE1 as the optimal geometry, the full-scale IREC competition vehicle is being designed in OnShape. The goal of this phase is to translate the CFD-optimized subscale geometry into dimensioned, manufacturable components. This section primarily showcases the CAD work completed to date— full vehicle assembly, component modeling, and engineering drawings for the fin assembly and boattail.

Vehicle Assembly

Fig. 17 — Exploded assembly BTTE1.

Fig. 18 — Assembled side profile BTTE1.

Fin Assembly — Engineering Drawing & Dimensions

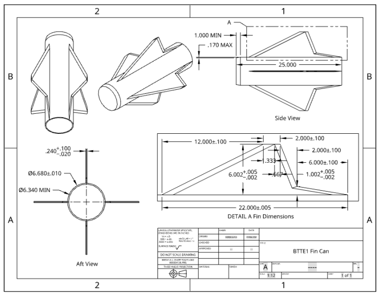

Fig. 19 — Fin assembly engineering drawing

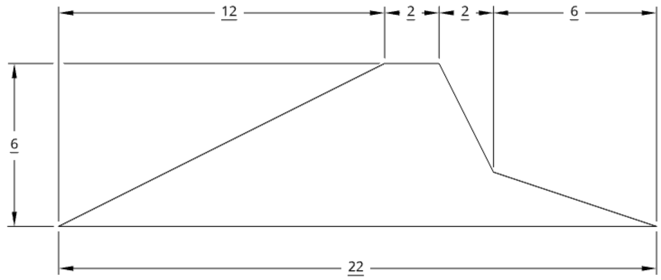

Fig. 20 — Fin planform dimensions.

Boattail — Engineering Drawing & Dimensions

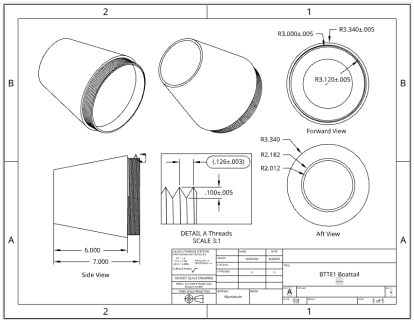

Fig. 21 — Boattail engineering drawing

Overview

The full scale design involves a forward, central and aft airframe for dual-deployment recovery. To test different motors, with different aft airframe configurations, a detachable fin can is designed to slide on and be constrained to the aft end of the aft airframe. The boattail is also removable with threads that will screw into the aft airframe. Structural shear and normal-force loading requirements will be finalized following higher-fidelity CFD analysis of the full-scale configuration.

Portfolio prepared May 2026 | Wind tunnel data, higher-fidelity CFD, and updated drawings to be added as work progresses. Research poster and CAD files available upon request.